

6. Nierówności kwadratowe 3

Rozwiąż nierówność:

a) $(2x-2)(x-3) < (x-3)(x-4)$

b) $(3x-1)(x+2) \geq (x-3)(2x-1)$

c) $(2x-2)^2 \geq (3x-2)(x-3)$

a) $(2x-2)(x-3) < (x-3)(x-4) \quad | :-(x-3)(x-4)$

$(2x-2)(x-3) - (x-3)(x-4) < 0$ *grupujemy wyrazy*

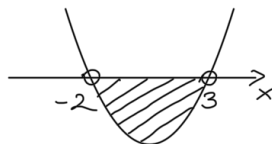
$(x-3)[2x-2 - (x-4)] < 0$

$(x-3)(2x-2-x+4) < 0$

$(x-3)(x+2) < 0$

$x-3=0$
 $x=3$

$x+2=0$
 $x=-2$



$x \in (-2, 3)$

b) $(3x-1)(x+2) \geq (x-3)(2x-1)$

W tym przypadku musimy wykonać mnożenie nawiasów

$3x^2 + 6x - x - 2 \geq 2x^2 - x - 6x + 3 \quad | -2x^2 + 7x - 3$

$x^2 + 12x - 5 \geq 0$

$\Delta = b^2 - 4ac$

$\Delta = 12^2 - 4 \cdot 1 \cdot (-5) = 144 + 20 = 164$

$\sqrt{\Delta} = \sqrt{164} = 2\sqrt{41}$

$x_1 = \frac{-b - \sqrt{\Delta}}{2a} = \frac{-12 - 2\sqrt{41}}{2} = -6 - \sqrt{41}$

$x_2 = \frac{-b + \sqrt{\Delta}}{2a} = \frac{-12 + 2\sqrt{41}}{2} = -6 + \sqrt{41}$



Odp: $x \in (-\infty; -6 - \sqrt{41}] \cup [-6 + \sqrt{41}; +\infty)$

c) $(2x-2)^2 \geq (3x-2)(x-3)$

$(a-b)^2 = a^2 - 2ab + b^2$

$4x^2 - 8x + 4 \geq 3x^2 - 9x - 2x + 6 \quad | -3x^2 + 11x - 6$
 $x^2 + 3x - 2 \geq 0$

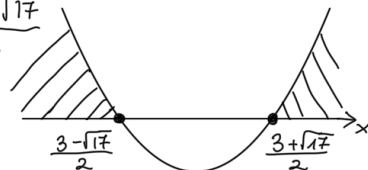
$\Delta = b^2 - 4ac$

$\Delta = 3^2 - 4 \cdot 1 \cdot (-2) = 9 + 8 = 17$

$\sqrt{\Delta} = \sqrt{17}$

$x_1 = \frac{-b - \sqrt{\Delta}}{2a} = \frac{-3 - \sqrt{17}}{2}$

$x_2 = \frac{-b + \sqrt{\Delta}}{2a} = \frac{-3 + \sqrt{17}}{2}$



Odp: $x \in (-\infty; \frac{-3 - \sqrt{17}}{2}] \cup [\frac{-3 + \sqrt{17}}{2}; +\infty)$